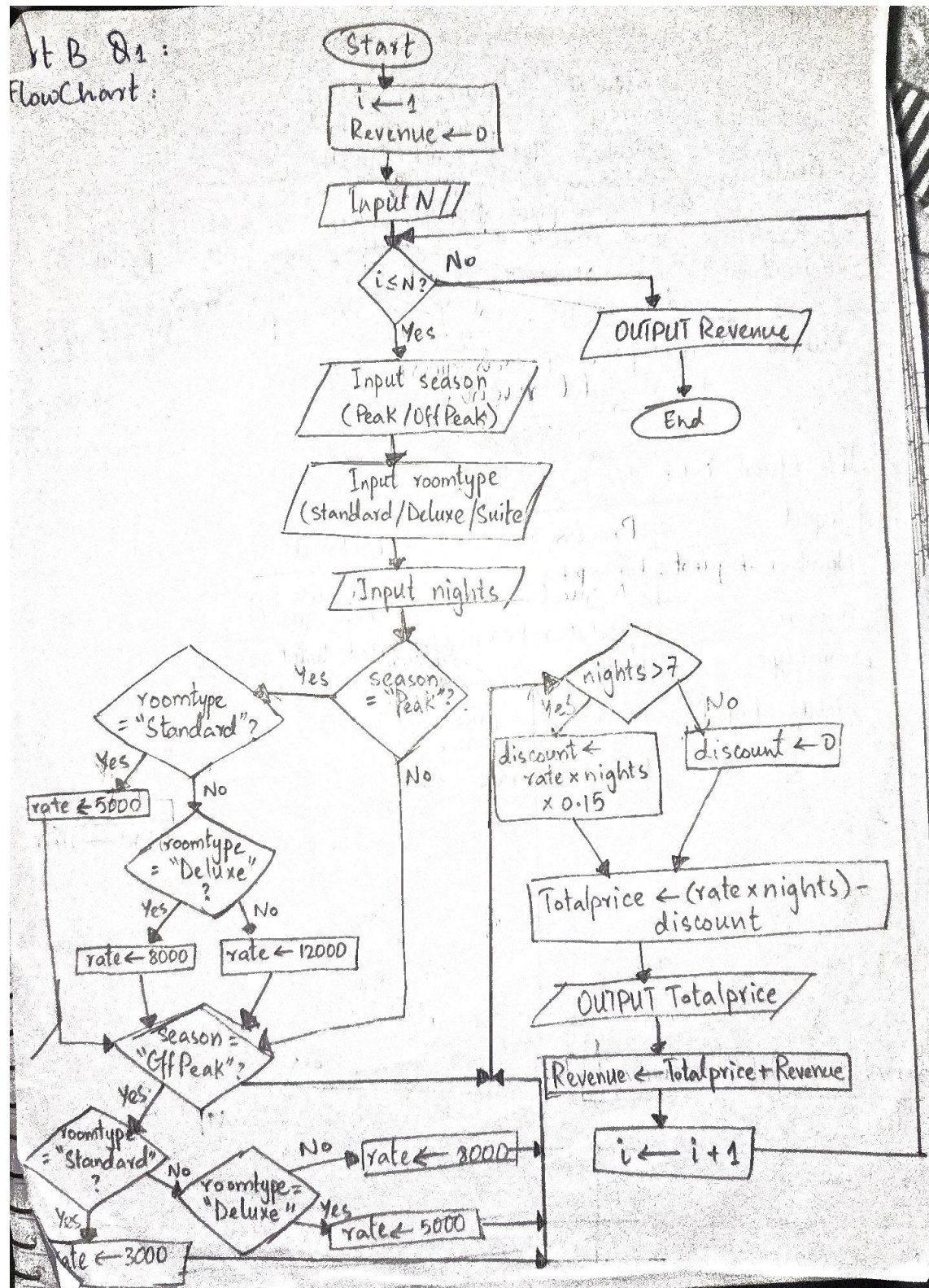


Q1

Flowchart:



Pseudocode:

START

Declare ~~n~~ rates : INTEGER

Declare nights : INTEGER

Declare discount : FLOAT

Declare Totalprice : FLOAT

Declare Revenue : FLOAT

Declare season : STRING

Declare roomtype : STRING

~~Declare discountamount : FLOAT~~

FOR $\leftarrow 1$ to N

season $\leftarrow$ Input "Enter Peak or OffPeak for Season: "

roomtype $\leftarrow$ Input "Enter Standard, Deluxe or Suite: "

nights $\leftarrow$ Input "Enter number of nights you are staying: "

~~IF nights $\geq$ 7 THEN~~

IF season = "Peak" THEN

IF roomtype = "Standard" THEN

rate $\leftarrow$ 5000

Endif

ELSE IF roomtype = "Deluxe" THEN

rate $\leftarrow$ 8000

Endif

ELSE

rate $\leftarrow$ 12000

Endif

IF season = "OffPeak" THEN

IF roomtype = "Standard" THEN

rate $\leftarrow$ 3000

Endif

ELSE IF roomtype = "Deluxe" THEN

rate $\leftarrow$ 5000

Endif

ELSE

rate $\leftarrow$ 8000

Endif

IF nights > 7 THEN
discount $\leftarrow$ (rate * nights) * 0.15

ELSE

discount $\leftarrow$ 0

Totalprice $\leftarrow$ (rate * nights) - discount

OUTPUT "Totalprice"

Revenue $\leftarrow$ Totalprice + Revenue

EndFOR

OUTPUT Revenue

END

IPO and PAC chart:

PAC Chart Q1:

Problem	Calculate the price for each guest based on season, roomtype and number of nights stayed, And then calculate total hotel revenue.
Analysis	Calculating ^{Determining} the room rate using season and room type and base price. Applying 15% discount if nights stayed is more than 7. Adding final price to Revenue.
Control	Loop for N guests, nested if else for season and roomtype and if for discount or on long stay
Output	Final price of each guest stay and total hotel revenue.

IPO Chart Q1:

Input	Process	Output
Number of guests	For loop for N guests	Final price for each guest
season	Calculate rate x nights	total hotel revenue
roomtype	Applying 15% discount	
nights stayed	Adding guest price to total revenue	
	Determine rate	